Q No1. What is Laplace transform why does kernal use?

Solution.

$$F(s) = L(f(t)) = \int_0^\infty e^{-st} f(t)\,dt.$$

$K(s,t) = e^{-st}$ is know as Kernal. By uses of Kernal:-

i) Easy to integrate or differentiate

2) By kernal use, differential equation into algebric equation.

3) Kernal smooth out the given function like sharp, edges, corners etc.

Q No2 Give Counter example of Existence Theorem;

Sol

$$f(t) = \frac{1}{\sqrt{t}}$$

As $t \to 0$ Then $f(t) \to \infty$

function on $[0, \infty)$. Also

$$|f(t)| \leq me^{kt} \qquad \text{if } m=1, k=0$$

Applying Laplace transform on given function

$$L(f(t)) = \int_0^\infty e^{-st} \frac{1}{\sqrt{t}}\, dt$$

Put $-st = -x^2$

$$\frac{1}{\sqrt{t}}\, dt = \frac{2}{\sqrt{s}}\, dx$$

$$L(f(t)) = \int_0^\infty e^{-x^2} \cdot \frac{2}{\sqrt{s}}\, dx$$

$$= \frac{2}{\sqrt{s}} \int_0^\infty e^{-x^2}\, dx$$

$$\int_0^\infty e^{-x^2} = \frac{\sqrt{\pi}}{2}$$

(Guassian integral)

$$L(f(t)) = \frac{2}{\sqrt{s}} \cdot \frac{\sqrt{\pi}}{2}$$

$$L(f(t)) = \sqrt{\frac{\pi}{s}} \qquad s>0$$

Hence $\frac{1}{\sqrt{t}}$ is not Piecewise Continous function.

But its Laplace transformation exist.

Q No3. Find Laplace Transform of Hyperbolic Function $L(\sinh at)$.

Sol

$$L(\sinh at) = L\left[\frac{1}{2}(e^{at} - e^{-at})\right]$$

$$= \frac{1}{2}\left[L(e^{at}) - L(e^{-at})\right]$$

$$= \frac{1}{2}\left[\frac{1}{s-a} - \frac{1}{s+a}\right] \Rightarrow \frac{1}{2}\left[\frac{s+a-s+a}{s^2-a^2}\right]$$

$$= \frac{1}{2}\left[\frac{2a}{s^2-a^2}\right] \Rightarrow \frac{a^2}{s^2-a^2}$$

$$L(\sinh at) = \frac{a^2}{s^2-a^2}$$

Q No4 Laplace Transform of nth Derivatives.

$$L(f^n) = s^n L(f) - s^{n-1} f(0) - s^{n-2} f(0) \cdots s^0 f^{n-1}(0)$$

CamScanner

Q.No5 Using Convolution Properties Prove that

$f * 1 \neq f$

Sol

Let $f(t) = t$

$$f(t) * 1 = \int_0^t f(\tau) \cdot 1 \, d\tau$$

$$= \int_0^t \tau \, d\tau \Rightarrow \frac{1}{2}\left[\tau^2\right]_0^t$$

$$= \frac{1}{2}\left[t^2 - 0\right] = \frac{t^2}{2}$$

$$= \frac{1}{2}t^2 \neq t = f(t)$$

Q.No6 $L\left(\frac{\partial u}{\partial t}\right) = s\tilde{u}(x,s) - u(x,0)$

Sol

$$\int_0^\infty e^{-st}\left(\frac{\partial u}{\partial t}\right) dt$$

$$= e^{-st}\left[u(x,t)\right]_0^\infty - (-s)\int_0^\infty u(x,t)\, e^{-st}\, dt$$

$$= 0 - 1 \cdot u(x,t) + sL\left(u(x,t)\right)$$

$$= s\tilde{u}(x,t) - u(x,0)$$

$$L\left(\frac{\partial u}{\partial t}\right) = s\tilde{u} - u(x,0)$$

NO 7.

Dirac's Delta Functions

$\int_0^\infty s(t-a) = 1$ Prove that.

Sol Let $f_k(t) = \begin{cases} 1/k & \text{if } 0 \le t \le k \\ 0 & \text{if otherwise} \end{cases}$

In general, $f_k(t-a) = \begin{cases} 1/k & \text{if } 0 \le t \le a+k \\ 0 & \text{if otherwise} \end{cases}$

$k \to 0$ Then

$$T_k = \lim_{k\to 0} f_k(t-a) = s(t-a) = \begin{cases} \infty & \text{if } t=a \\ 0 & \text{if otherwise} \end{cases}$$

Such That

$$\text{Integral} = \int_0^\infty s(t-a) = 1$$

Example :-

$$\int_a^{a+k} f_k(t-a)\,dt$$

$$= \int_0^{a+k} \frac{1}{k}\,dt \Rightarrow \frac{1}{k}\left[t\right]_a^{a+k}$$

$$= \frac{1}{k}\left[a+k-a\right] \Rightarrow \frac{1}{k}\left[k\right]$$

$$= \frac{k}{k} = 1$$

CamScanner

Q: Laplace Transform of 1st derivatives

So' $L(f') = sL(f) - f(0)$

f is differentiable $\Rightarrow$ f is continuous

$$L(f') = L(f'(t)) = \int_0^\infty e^{-st} f'(t)\,dt$$

Integrat and applying limits we have.

$$L(f') = \left| e^{-st} f(t) \right|_0^\infty - \int_0^\infty f(t)(-s)e^{-st}\,dt$$

$$L(f') = \left| \text{Limits } t\to\infty \ \frac{f(t)}{e^{st}} - \frac{f(0)}{e^0} \right| + s\int_0^\infty e^{-st} f(t)\,dt$$

$$L(f') = 0 - f(0) + sL(f)$$

$$L(f') = sL(f) - f(0)$$

2nd derivative :-

$$L(f'') = sL(f') - f'(0)$$

$$= s[sL(f) - f(0)] - f'(0)$$

$$= [s^2 L(f) - sf(0)] - f'(0)$$

$$L(f'') = s^2 L(f) - sf(0) - f'(0)$$

No9 Laplace Transform of Rodriguez Formula of Laguerre's Polynomial:-

$$L_n(t) = \frac{e^t}{n!}\frac{d^n}{dt^n}(t^n e^{-t})$$

Applying Laplace Transform.

$$L\{L_n(t)\} = L\left\{\frac{e^t}{n!}\frac{d^n}{dt^n}(t^n e^{-t})\right\}$$

$$L\{t^n\} = \frac{n!}{s^{n+1}}$$

$$L\{e^{-t}t^n\} = \frac{n!}{(s+1)^{n+1}}$$

$$L\left\{\frac{d^n}{dt^n}(e^{-t}t^n)\right\} = s^n L\{e^{-t}t^n\} = \frac{s^n n!}{(s+1)^{n+1}}$$

$F(s) \to F(s-1)$

$$L\left\{e^t\frac{d^n}{dt^n}(e^{-t}t^n)\right\} = n!\frac{(s-1)^n}{s^{n+1}}$$

$$\frac{1}{n!}L\left\{e^t\frac{d^n}{dt^n}(e^{-t}t^n)\right\} = \frac{(s-1)^n}{s^{n+1}}$$

$$L\left\{\frac{e^t}{n!}\frac{d^n}{dt^n}(e^{-t}t^n)\right\} = \frac{(s-1)^n}{s^{n+1}}$$

This is our Required result.

Q No 10 Prove That $L^{-1}(H(s)) = h(t)$ where

$$H(s) = \frac{1}{(s-a)s}$$

Solution

if $L(f) = F(s)$, $L(g) = G(s)$

$$F(s)G(s) = L\left\{\int_0^t f(\tau)\, g(t-\tau)\, d\tau\right\}$$

$$L^{-1}\left(F(s)\,G(s)\right) = \int_0^t f(\tau)\, g(t-\tau)\, d\tau$$

Given That $H(s) = \dfrac{1}{(s-a)s} = F(s)\,G(s)$

using result

$\therefore F(s) = \frac{1}{s-a} \Rightarrow f(t)$

$L^{-1}F(s) = L^{-1}\left(\frac{1}{s-a}\right) = e^{at} \Rightarrow f(t)$

$L^{-1}F(s) = e^{at} \Rightarrow f(\tau)$

$\therefore G(s) = \frac{1}{s} \Rightarrow g(t)$

$L^{-1}G(s) = L^{-1}\left(\frac{1}{s}\right) = 1 \Rightarrow g(t-\tau)$

$L^{-1}G(s) = 1 \Rightarrow g(t-\tau)$

$$L^{-1}H(s) = L^{-1}\left(F(s)\,G(s)\right)$$

$$= \int_0^t e^{at} \cdot 1 \, d\tau$$

Integrate and applying limits we have.

$$L^{-1}(H(s)) = \frac{1}{a}\left[e^{at}\right]_0^t \Rightarrow \frac{1}{a}\left[e^{at} - 1\right]$$

$$L^{-1}(H(s)) = h(t)$$

CamScanner

Q No 11

Unit step Function

$$u(t-a) = \begin{cases} 1 & \text{if } t > 0 \\ 0 & t < a \end{cases}$$

Sol

Applying Laplace Transform.

$$L(u(t-a)) = \int_0^\infty e^{-st}\, u(t-a)\, dt$$

$$= \int_0^a e^{-st}(0)\, dt + \int_a^\infty e^{-st}(1)\, dt \qquad a > 0$$

$$= 0 + \frac{1}{-s}\left| e^{-st} \right|_a^\infty$$

$$= -\frac{1}{s}(0 - e^{-sa}) \Rightarrow \frac{1}{s}(e^{-as})$$

$$L(u(t-a)) = \frac{1}{s} e^{-as}$$

This is our Required result.

Q No 12

Volterra Integral Equation

The type of equation that involves unknown function which appears inside and outside an integral. For example

$$y(t) - \int_0^t y(t) \sin(t-\tau)\, dt = t$$

Q No 13 Volterra Integral Equation of 2nd kind

$$y(t) - \int_0^t (1+t)\, y(t-\tau)\, d\tau = 1 - \sinh t$$

Q No 14 $L(t \sin wt)$

Sol

$$f(t) = t \sin wt \Rightarrow f(0) = 0$$

$$f'(t) = t \cos wt \cdot w + 1 \sin wt \Rightarrow f'(0) = 0$$

$$f''(t) = w[t - w \sin wt + \cos wt] + w \cos wt$$

$$f''(t) = -w^2(t \sin wt) + w \cos wt + w \cos wt$$

$$= -w^2 t \sin wt + 2w \cos wt$$

We know that

$$L(f'') = s^2 L(f) - s f'(0) - f'(0)$$

By Pulling values

$$L\{-w^2 t \sin wt + 2w \cos wt\} = s^2 L(f) - 0 - 0$$

$$-w^2 L(t \sin wt) + 2w L(\cos wt) = s^2 L(f)$$

$\because \cos wt = \frac{s}{s^2+w^2}$

$$-w^2 L(f) + 2w L\left(\frac{s}{s^2+w^2}\right) = s^2 L(f)$$

$$\frac{2ws}{s^2+w^2} = L(f)(s^2+w^2) \Rightarrow L(f) = \frac{2ws}{(s^2+w^2)^2}$$

CamScanner

Q No 00

$$L^{-1}\ln\left\{\ln\left(\frac{s+b}{s+a}\right)^{-s}\right\} \qquad a, b > 0$$

Sol

$$\ln\left(\frac{s+b}{s+a}\right)^{-s^{-1}} \Rightarrow \ln\left(\frac{s+b}{s+a}\right)^{-\frac{1}{s}}$$

$$\ln\left(\frac{s+a}{s+b}\right)^{\frac{1}{s}} \Rightarrow \frac{1}{s}\ln\left(\frac{s+a}{s+b}\right) \quad \text{---} \quad (i)$$

Now

$F(s) = \ln\left(\frac{s+a}{s+b}\right) = \ln(s+a) - \ln(s+b)$

$F(s) = \ln(s+a) - \ln(s+b)$

$$F'(s) = \frac{1}{s+a} - \frac{1}{s+b}$$

Applying inverse Transform.

$$L^{-1}(F'(s)) = L^{-1}\left(\frac{1}{s+a}\right) - L^{-1}\left(\frac{1}{s+b}\right)$$

$$\because L^{-1}(F'(s)) = (-1)tf(t)$$

$$L^{-1}\left(\frac{1}{s+a}\right) = e^{-at}, \quad L^{-1}\left(\frac{1}{s+b}\right) = e^{-bt}$$

$$(-1)tf(t) = e^{-at} - e^{-bt} \Rightarrow f(t) = -\frac{e^{-at} - e^{-bt}}{t}$$

$$f(t) = \frac{e^{-bt} - e^{-at}}{t}$$

Hence

$$L^{-1}\left\{\frac{F(s)}{s}\right\} = \int_0^t f(s)\,ds = \int_0^t \frac{e^{-bs} - e^{-as}}{s}\,ds$$

This is

Q No 16

Laplace Transform of t^n $(n \in N)$

Prove that $L(t^n) = \frac{n!}{s^{n+1}}$ $n \in N$

Sol: we will Prove by mathematical induction

For $n = 0$

$$L(t^0) = \frac{0!}{s^{0+1}}$$

$$L(1) = \frac{1}{s}$$

For $n+1$

$$L(t^{n+1}) = \frac{(n+1)!}{s^{n+2}}$$

Applying Laplace Transform

$$L(t^{n+1}) = \int_0^\infty e^{-st}(t^{n+1})\,dt$$

integrate and applying limits we have.

$$L(t^{n+1}) = \left| t^{n+1}\left(\frac{e^{-st}}{-s}\right)\right|_0^\infty - \int_0^\infty \left(\frac{e^{-st}}{-s}\right)(n+1)t^n\,dt$$

$$= -\frac{1}{s}\left[\frac{t^{n+1}}{e^{st}}\right]_0^\infty - \left(\frac{n+1}{-s}\right)\int_0^\infty e^{-st}t^n\,dt$$

$$= -\frac{1}{s}\left[\frac{t^{n+1}}{e^{st}}\right]_0^\infty + \frac{n+1}{s}L(t^n) \qquad \because \int_0^\infty e^{-st}t^n dt = L(t^n)$$

$$= \frac{-1}{s}\left[\frac{t^{\infty+1}}{e^{\infty}} - 0\right] + \frac{n+1}{s}L(t^n)$$

$$= \frac{-1}{s}[0-0] + \frac{n+1}{s}L(t^n)$$

CamScanner

$$L(t^{n+1}) = 0 + \frac{n+1}{s} L(t^n)$$

$$L(t^{n+1}) = \frac{n+1}{s} L(t^n)$$

$$\therefore L(t^n) = \frac{n!}{s^{n+1}}$$

$$L(t^{n+1}) = \frac{n+1}{s} \frac{n!}{s^{n+1}}$$

$$L(t^{n+1}) = \frac{(n+1)!}{s^{n+2}}$$

It is true for $n+1$. Hence Proved.

Q No17 Find $L(\sin wt)$ by using derivative expression of Laplace.

Sol

$$f(t) = \sin wt \qquad f(0) = 0$$

$$f'(t) = w \cos wt \qquad f'(0) = w$$

$$f''(t) = -w^2 \sin wt$$

Putting the values

$$L(f'') = s^2 L(f) - s f(0) - f'(0)$$

$$-w^2 \sin wt = s^2 L(f) - 0 - w$$

$$-w^2 L(\sin wt) = s^2 L(f) - w$$

$$-w^2 L(f) = s^2 L(f) - w$$

$$L(f) = \frac{w}{s^2 + w^2}$$

CamScanner

NO16

S- Shifting Theorem

If $f(t)$ has the transform $F(s)$ where $s > k$

Then $e^{at} f(t)$ has the transform $F(s-a)$ where $(s-a) > k$.

$$L\{e^{at} f(t)\} = F(s-a)$$

In term of inverse Laplace transform

$$L^{-1}\{F(s-a)\} = e^{at} f(t)$$

Proof As we know Laplace Transform

$$L(f(t)) = F(s) = \int_0^\infty e^{-st} f(t)\,dt$$

For given function

$$F(s-a) = \int_0^\infty e^{-(s-a)t} f(t)\,dt$$

$$\int_0^\infty e^{-st} e^{at} f(t)\,dt \Rightarrow \int_0^\infty e^{-st} (e^{at} f(t))\,dt$$

$$F(s-\alpha) = L(e^{\alpha t} f(t))$$

$$L^{-1}(F(s-\alpha)) = e^{\alpha t} f(t)$$

This is our Required result.

Find $L(\sin at)$

$$F_s = \int_0^\infty e^{-st}(\sin at)\,dt$$

$$= \frac{e^{-st}}{-s}\sin at\Big|_0^\infty - \frac{1}{-s}\int_0^\infty e^{-st}(a\cos at)\,dt$$

$$= \frac{1}{-s}\left[\frac{\sin at}{e^{st}}\right]_0^\infty + \frac{a}{s}\int_0^\infty e^{-st}\cos at\,dt$$

$$= \frac{1}{-s}\left[\frac{\sin\infty}{e^\infty} - 0\right] + \frac{a}{s} \qquad \therefore F_c = \int_0^\infty e^{-st}\cos at\,dt$$

$$= \frac{-1}{s}[0-0] + \frac{a}{s}\cdot F_c$$

$$F_s = \frac{a}{s}F_c$$

$$F_c = L(\cos at) = \frac{s}{s^2+a^2}$$

$$F_s = \frac{a}{s}F_c$$

$$= \frac{a}{s}\cdot\frac{s}{s^2+a^2}$$

$$F_s = \frac{a}{s^2+a^2}$$

$$F_s = L(\sin at)$$

$$L(\sin at) = \frac{a}{s^2+a^2}$$

Laplace Transform t^α $(\alpha > -1)$

Sol

$$L(t^\alpha) = \int_0^\infty e^{-st} t^\alpha \, dt$$

$$L(t^\alpha) = \int_0^\infty e^{-v} \left(\frac{v}{s}\right)^\alpha \frac{1}{s} \, dv$$

$\therefore v = st$

$dt = \frac{1}{s} dv$

$t = \frac{v}{s}$

$$L(t^\alpha) = \int_0^\infty e^{-v} \left(\frac{v^\alpha}{s^\alpha}\right) \frac{1}{s} \, dv$$

$$= \int_0^\infty e^{-v} \frac{v^\alpha}{s^{\alpha+1}} \, dv \Rightarrow \frac{1}{s^{\alpha+1}} \int_0^\infty e^{-v} v^\alpha \, dv$$

$$\int_0^\infty e^{-v} v^\alpha \, dv = \text{Gamma Function}$$

$$\Gamma(\alpha+1)$$

$$L(t^\alpha) = \frac{1}{s^{\alpha+1}} \left(\Gamma(\alpha+1)\right)$$

$$L(t^\alpha) = \frac{\Gamma(\alpha+1)}{s^{\alpha+1}}$$

This is our required result